

5.3 Principles of Electrochemistry (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.3 Principles of Electrochemistry (A Level Only)
Difficulty	Hard

Time allowed: 40

Score: /28

Percentage: /100

Question 1a

Fig. 1.1 shows a lithium–iodine electrochemical cell. These cells are often used in pacemakers as they are reliable and have a life span in the region of 10 years.

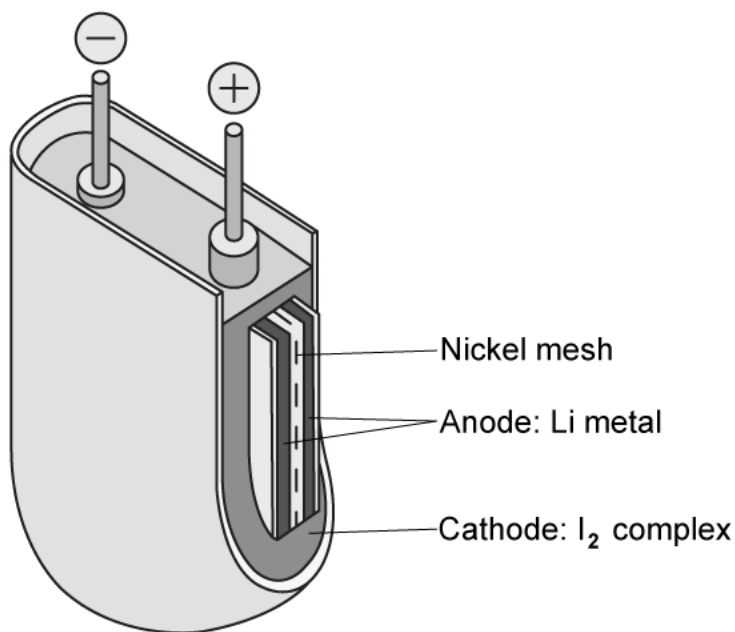


Fig. 1.1

It consists of a lithium electrode and an inert electrode immersed in body fluids that are separated by a nickel mesh and collect charge from the anode. It has a high internal resistance which means that only a low current can be drawn.

Explain why the lithium–iodine electrochemical cell is a **dry** cell.

[2 marks]

Question 1b

Write the overall equation for the reaction taking place at the electrodes of the lithium–iodine electrochemical cell when a current flows.

- overall equation

[1 mark]

Question 1c

Table 1.1 lists electrode potentials for some electrode reactions.

Table 1.1

Electrode reaction	$E^\ominus / \text{V}$
$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+ 0.54
$\text{Li}^+ + \text{e}^- \rightleftharpoons \text{Li}$	- 3.04
$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$	- 0.25

Calculate the $E^\ominus_{\text{cell}}$ for the lithium-iodine electrochemical cell.

$$E^\ominus_{\text{cell}} = \dots\dots\dots \text{V}$$

[1 mark]

Question 1d

A current of $1.5 \times 10^{-5} \text{ A}$ is drawn from this cell.

Calculate the number of days for 0.08 g of the lithium electrode to be used up. Assume the current remains constant throughout this period. Show your working.

The Faraday constant, $F = 9.65 \times 10^4 \text{ C mol}^{-1}$.

$$\text{time} = \dots\dots\dots \text{days}$$

[4 marks]

Question 2a

A student set up an electrochemical cell using a concentrated sodium chloride solution using a current of 6 A.

State the half-equations occurring at the electrodes during the electrolysis of the concentrated aqueous solution of sodium chloride.

Cathode

Anode

[2 marks]**Question 2b**

Calculate the time, in minutes, to produce 2.00 dm^3 of gas at the **anode** at standard temperature and pressure.

The Faraday constant, $F = 9.65 \times 10^4 \text{ C mol}^{-1}$.

State your answer to 2 significant figures and show your working.

time = minutes

[3 marks]**Question 2c**

The student changed the electrolyte to a very dilute sodium chloride solution.

State what change would occur at the anode and give the half equation for the process.

[2 marks]

Question 2d

In a different electrolysis experiment, copper(II) sulfate solution was electrolysed using graphite electrodes.

Table 2.1 lists electrode potentials for some electrode reactions.

Table 2.1

Electrode reaction	$E^{\ominus} / \text{V}$
$\text{Cu}^{+} + \text{e}^{-} \rightleftharpoons \text{Cu}$	+ 0.52
$\text{Cu}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Cu}$	+ 0.34
$\text{Cu}^{2+} + \text{e}^{-} \rightleftharpoons \text{Cu}^{+}$	+ 0.15
$2\text{H}_2\text{O} + 2\text{e}^{-} \rightleftharpoons \text{H}_2 + 4\text{OH}^{-}$	- 0.83
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^{-} \rightleftharpoons 4\text{OH}^{-}$	+ 0.40
$\frac{1}{2}\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{O}$	+ 1.23
$\text{SO}_4^{2-} + 4\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{SO}_2 + 2\text{H}_2\text{O}$	+ 0.17
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^{-} \rightleftharpoons 2\text{SO}_4^{2-}$	+ 2.01

Explain how the products at the anode and cathode are produced.

[5 marks]

Question 3a

Explain why Fig. 3.1 does not represent the standard hydrogen electrode.

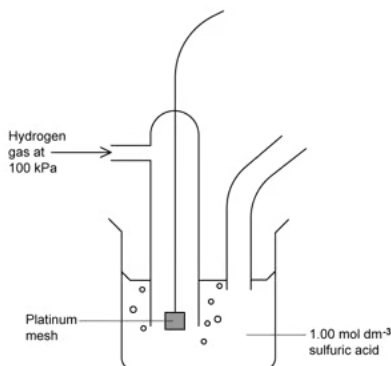


Fig. 3.1

[2 marks]

Question 3b

The standard electrode potential for $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Zn}(\text{s})$ is -0.76 V .

State the meaning of the minus sign in the value of -0.76 V .

[1 mark]

Question 3c

Zinc coating on metals serves as physical protection which prevents rust from affecting the underlying metal surface. This is achieved by electroplating as shown in Fig. 3.2.

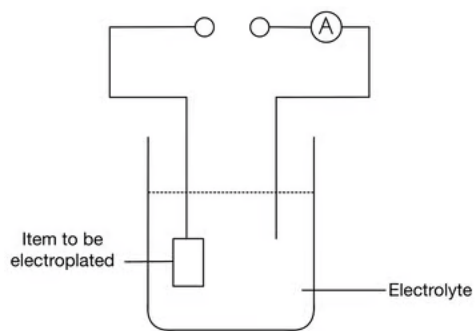


Fig. 3.2

- i)
Suggest a suitable solution to act as the electrolyte during zinc electroplating.

[1]

- ii)
Complete the diagram by labelling the polarity of the power source by using a + and - sign.

[1]

[2 marks]

Question 3d

Calculate the length of time, in hours, required to deposit 1.0 g of zinc on the item to be electroplated. Assume the current is a constant 0.1 A throughout this period.

The Faraday constant, $F = 9.65 \times 10^4 \text{ C mol}^{-1}$.

State your answer to 2 significant figures and show your working.

time = hours

[3 marks]

Model Answers: Hard

1a

a) The lithium-iodine electrochemical cell is a **dry** cell because:

- There is no water / liquid (in the electrochemical cell); [1 mark]
- Water / liquid could react with the lithium electrode

AND

Which could cause the battery / cell to explode **OR** which could cause a build-up in pressure (inside the cell); [1 mark]

[Total: 2 marks]

- This question incorporates your knowledge of the reactivity of Group 1 metals
 - Group 1 metals react with water to form a metal hydroxide solution and flammable hydrogen gas
- Since the lithium-iodine electrochemical cell is a sealed container, the production of hydrogen gas would cause a pressure build-up that could result in the cell exploding

1b

b) The overall equation for the reaction taking place at the electrodes of the lithium-iodine electrochemical cell, when a current flows is:

- $2\text{Li} + \text{I}_2 \rightarrow 2\text{Li}^+ + 2\text{I}^-$; [1 mark]

[Total: 1 mark]

- The information in this question, as well as part (a), gives you the reactants for the two half-equations
 - Lithium
 - Iodine
- This means that:
 - Lithium will be oxidised to lithium ions
 - $\text{Li} \rightarrow \text{Li}^+ + \text{e}^-$

- Iodine will be reduced to iodide ions
 - $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$
- **Remember:** Combining half-equations to form an overall equation requires both half-equations to have the same number of electrons
 - This means that the two half-equations required to form the overall equation are:
 - $2\text{Li} \rightarrow 2\text{Li}^+ + 2\text{e}^-$
 - $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$
 - These combine to give:
 - $2\text{Li} + \text{I}_2 + 2\text{e}^- \rightarrow 2\text{Li}^+ + 2\text{I}^- + 2\text{e}^-$
 - Cancelling the electrons gives the correct overall equation:
 - $2\text{Li} + \text{I}_2 \rightarrow 2\text{Li}^+ + 2\text{I}^-$

1c

c) $E^\ominus_{\text{cell}}$ for the lithium-iodine electrochemical cell is:

- +3.58 V; [1 mark]

[Total: 1 mark]

- **Remember:** $E^\ominus_{\text{cell}} = E^\ominus_{\text{reduction}} - E^\ominus_{\text{oxidation}}$
- The two relevant equations are:
 - $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$
 - $\text{Li}^+ + \text{e}^- \rightleftharpoons \text{Li}$
- The reduction equation has the higher / more positive $E^\ominus_{\text{cell}}$ value
 - Therefore, the calculation is $E^\ominus = (+0.54) - (-3.04) = +3.58 \text{ V}$

1d

d) To calculate the number of days for 0.08 g of the lithium electrode to be used up:

- Moles of lithium = $\frac{0.08}{6.9} = 1.16 \times 10^{-2}$; [1 mark]
- $Q = 1119$ (**OR** 1118.84...) C; [1 mark]
- $t = 7.46 \times 10^7 \text{ s}$; [1 mark]
- $t = 863 \text{ days}$; [1 mark]

[Total: 4 marks]

- Calculate the number of moles of lithium
 - $\frac{0.08}{6.9} = 1.16 \times 10^{-2}$ moles
- One faraday is the amount of electric charge carried by 1 mole of electrons or 1 mole of singly charged ions
 - So, $Q = \text{moles} \times F$ For $Q = n \times F$
 - $Q = 96500 \times 1.16 \times 10^{-2} = 1119$ (**OR** 1118.84...) C
- Use $Q = I \times t$ to calculate the time in seconds
 - $t = \frac{1119}{1.5 \times 10^{-5}} = 7.46 \times 10^7$ s
- Convert t from seconds to days
 - $t = \frac{7.46 \times 10^7}{(60 \times 60 \times 24)} = 863$ days

2a

a) The half equations occurring at each electrode are:

- Cathode:
 $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
OR
 $\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2}\text{H}_2$
OR
 $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$; [1 mark]
- Anode:
 $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ / $2\text{Cl}^- - 2\text{e}^- \rightarrow \text{Cl}_2$
OR
 $\text{Cl}^- \rightarrow \frac{1}{2}\text{Cl}_2 + \text{e}^-$ / $\text{Cl}^- - \text{e}^- \rightarrow \frac{1}{2}\text{Cl}_2$; [1 mark]

[Total: 2 marks]

- One mark can be awarded if the half-equations are correct but placed at the wrong electrodes
- During electrolysis
 - The cathode is the negative electrode and reduction occurs
 - The anode is the positive electrode and oxidation occurs
- The electrolysis of concentrated sodium chloride differs from that of the electrolysis dilute sodium chloride solution and this is often poorly represented by students in exams
- When the concentration of the sodium chloride increases to more than 25% the

Cl^- becomes preferentially discharged and chlorine gas is the main product of the reaction at the **anode**

- The overall reaction equation is:
 - $2\text{NaCl}(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g}) + \text{Cl}_2(\text{g})$

2b

b) To calculate the time, in minutes, for the production of 2.00 dm^3 at the anode:

Chlorine gas produced

- Moles of chlorine = $\frac{2 \text{ dm}^3}{22.4 \text{ dm}^3 \text{ mol}^{-1}} = 0.09 \text{ (mol)}$

AND

moles of electrons = $0.09 \times 2 = 0.18 \text{ (mol)}$; [1 mark]

- Charge = $0.18 \times 96\,500 \text{ C mol}^{-1} = 17232.14 \text{ (C)}$; [1 mark]
- Time (s) = $\frac{17232.14 \text{ C}}{6 \text{ A}} = 2872.02 \text{ (s)}$

AND

Time (min) = $\frac{2872.02 \text{ s}}{60} = 48 \text{ (min)}$; [1 mark]

[Total: 3 marks]

- For this question, you need to apply and rearrange
 - $Q \text{ (C)} = I \text{ (A)} \times t \text{ (s)}$
- If the half-equation at the anode is:
 - $2\text{Cl}^-(\text{aq}) \rightarrow \text{Cl}_2(\text{g}) + 2\text{e}^-$ then 2 moles of electrons must be removed from 2 moles of chloride ions to produce 1 mole of chlorine gas
 - This is why we need to multiply the moles of chlorine by 2Q
- Quantity of electricity in Coulombs = current in A x time in **seconds**
 - $17232.14 = 6 \times \text{time in seconds}$, so time in seconds = $17232.14 / 6 = 2872.02$ seconds
- There are two hurdles given. which are the
 - Conversion of s to min
 - Stating the answer to 2 significant figures

2c

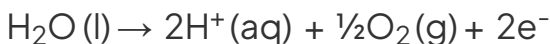
c) The change at the anode is:

- Oxygen gas / O_2 produced; [1 mark]

The half equation for dilute sodium chloride solution is:

- $2H_2O(l) \rightarrow 4H^+(aq) + O_2(g) + 4e^-$

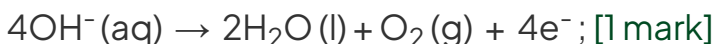
OR



OR



OR



[Total: 2 marks]

- At the anode, the following are possible processes:

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$	$E^\ominus = -1.36 \text{ V}$
$2H_2O(l) \rightarrow 4H^+(aq) + O_2(g) + 4e^-$	$E^\ominus = -1.23 \text{ V}$
$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$	$E^\ominus = -0.40 \text{ V}$

- If the sodium chloride solution is reasonably concentrated, you will get mainly chlorine
- If the sodium chloride solution is **very** dilute, you will get mainly oxygen
- At in-between concentrations, you will get a mixture of both

2d

d) The products at the anode and cathode are:

d) The products at the anode and cathode are:

- Anode: Oxygen gas / $O_2(g)$ **OR** Carbon dioxide / $CO_2(g)$
AND
Cathode: Copper metal / $Cu(s)$; [1 mark]

At the cathode:

- $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ $E^\ominus = +0.34 V$
AND
 $2H_2O(l) + 2e^- \rightarrow H_2(g) + 4OH^-(aq)$ $E^\ominus = -0.83 V$
AND
 $Cu^{2+}(aq) / Cu(s)$ preferably reduced as more positive $E^\ominus$; [1 mark]

At the anode:

- Sulfate ions can not be oxidised
AND
Hydroxide ions (from the solution) are oxidised to form water; [1 mark]
- $2H_2O(l) \rightarrow 4H^+(aq) + O_2(g) + 4e^-$; [1 mark]
- The oxygen produced reacts with the carbon / graphite (electrode) to form carbon dioxide; [1 mark]

[Total: 5 marks]

- The electrolysis of copper sulfate solution, $CuSO_4(aq)$ forms metallic copper
 - Therefore, the reduction reaction occurring at the cathode is $Cu^{2+} + 2e^- \rightleftharpoons Cu$ with an electrode potential of + 0.34V
- The electrode potential value for the Cu^{2+} ion means that:
 - Any equations with an electrode potential value greater than 0.34 V cannot be used
 - The $Cu^{2+} + 2e^- \rightleftharpoons Cu^+$ half-equation is not applicable
 - This is because copper metal is formed
 - The $SO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons SO_2 + 2H_2O$ half-equation is not applicable
 - This reaction would proceed backwards making SO_4^{2-} which opposes the electrolysis of copper(II) sulfate
 - The $2H_2O + 2e^- \rightleftharpoons H_2 + 4OH^-$ is applicable
 - This reaction can proceed in the backward direction, using the hydroxide ions (from the solution) to form water
- The water formed is used in a separate reaction to form oxygen which can react

with the carbon / graphite in the electrode to form carbon dioxide, which causes deterioration of the electrode

3a

a) Fig. 3.1 does not represent the standard hydrogen electrode because:

- Sulfuric acid is diprotic; [1 mark]
- The solution does not have a hydrogen ion concentration of 1.0 mol dm^{-3} ; [1 mark]

[Total: 2 marks]

- **Remember:** Standard conditions for electrochemistry are
 - 298 K
 - 100 kPa
 - 1.00 mol dm^{-3} ion concentrations
 - 1 mol dm^{-3} sulfuric acid does not actually give 2 mol dm^{-3} of hydrogen ions
 - This is why an answer stating that the solution contains $2.0 \text{ mol dm}^{-3} \text{ H}^+$ ions is not an accepted answer
 - Zero current
- The platinum electrode does **not** form part of the standard conditions
 - In some cases, platinum can be replaced by other metals such as rhodium, ruthenium and palladium (similar to the metals that can be used in catalytic converters)
- Sulfuric acid is a strong acid so the first dissociation of sulfuric acid is complete
 - $\text{H}_2\text{SO}_4 \rightarrow \text{H}^+ + \text{HSO}_4^-$
 - The HSO_4^- formed is a weak acid and doesn't fully dissociate
 - $\text{HSO}_4^- \rightleftharpoons \text{H}^+ + \text{SO}_4^{2-}$
 - Also, the hydrogen ions from the first dissociation will drive the equilibrium of the second dissociation backwards due to the increased hydrogen ion concentration

3b

b) The meaning of -0.76 V is:

- Electrons flow from the half-cell to the hydrogen electrode

OR

Zn is a better reducing agent than H_2

OR

Zn is oxidised more readily than H_2 ; [1 mark]

[Total: 1 mark]

- Explanations of standard electrode potentials and defining the standard hydrogen electrode (SHE) are not well answered
- Make sure you learn the conditions for the SHE
- Since the hydrogen electrode is always on the left by convention, the polarity of the half cell measured is always with respect to hydrogen
- The half-reaction will therefore always be a reduction reaction
- The more negative the value; the better the half-cell is at pushing electrons so the equilibrium lies to the left
- Therefore, for $Zn^{2+}(aq) + 2e^- \rightarrow Zn(s)$ electrons flow from the half-cell to the hydrogen electrode

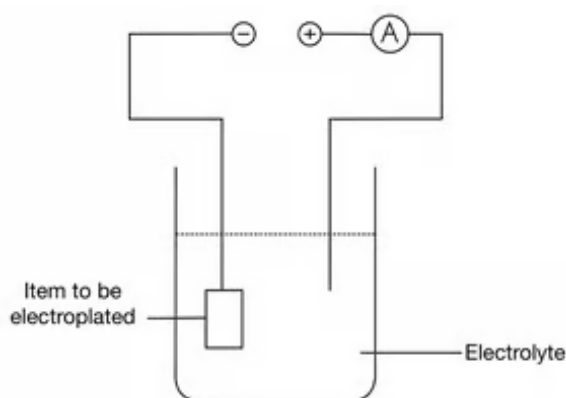
3c

c)

i) The electrolyte could be:

- Zinc sulfate / zinc chloride / zinc nitrate; [1 mark]

ii) The polarity of the cell is:



- Left hand side is '-'

AND

Right hand side is '+' ; [1 mark]

[Total: 2 marks]

- Examiners comment that students do not name possible electrolytes well for electroplating
- The electrolyte must be soluble and contain the metal ion of the metal being used to electroplate, so Zn^{2+} ions
- The anode will be made of zinc

3d

d) The length of time required to deposit 1 gram of zinc is:

- Moles of zinc = $\frac{1 \text{ g}}{65.4 \text{ g mol}^{-1}} = 0.015 \text{ (mol)}$

AND

moles of electrons = $0.015 \text{ mol} \times 2 = 0.031 \text{ (mol)}$; [1 mark]

- $Q = 0.031 \times 96\,500 = 2951.07 \text{ (C)}$; [1 mark]

- $t = \frac{2951.07 \text{ C}}{0.1 \text{ A}} = 29\,510.70 \text{ s}$

AND

$$t = \frac{29510.70 \text{ s}}{(60 \times 60)} = 8.2 \text{ hours}; [1 \text{ mark}]$$

[Total: 3 marks]

- **Remember:** The number of moles of zinc must be multiplied by 2 to determine the number of moles of electrons
 - $\text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Zn}$